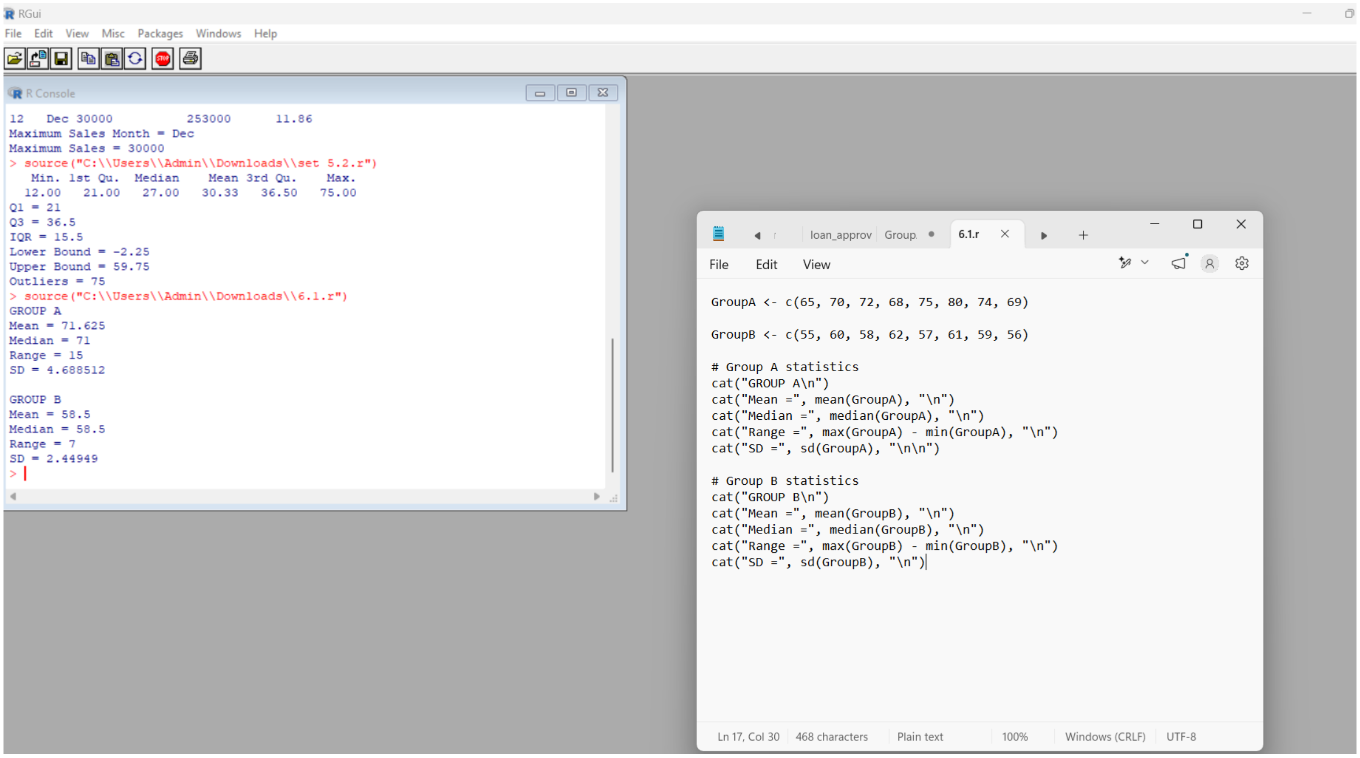
SET 6

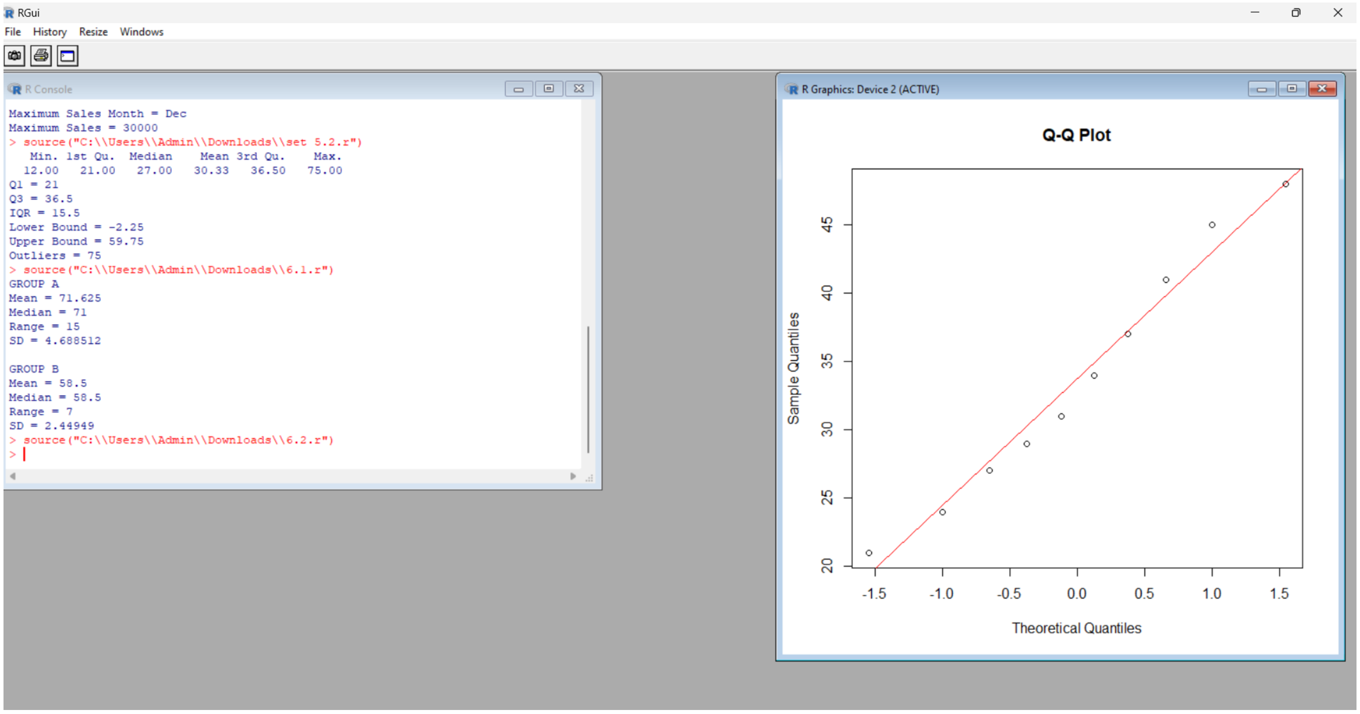
1. R: Create two groups of exam scores; calculate group-wise mean, median, range and SD and determine consistency.

Output:



2. R: Construct a Q-Q plot for 21,24,27,29,31,34,37,41,45,48 and comment on normality.

Output:

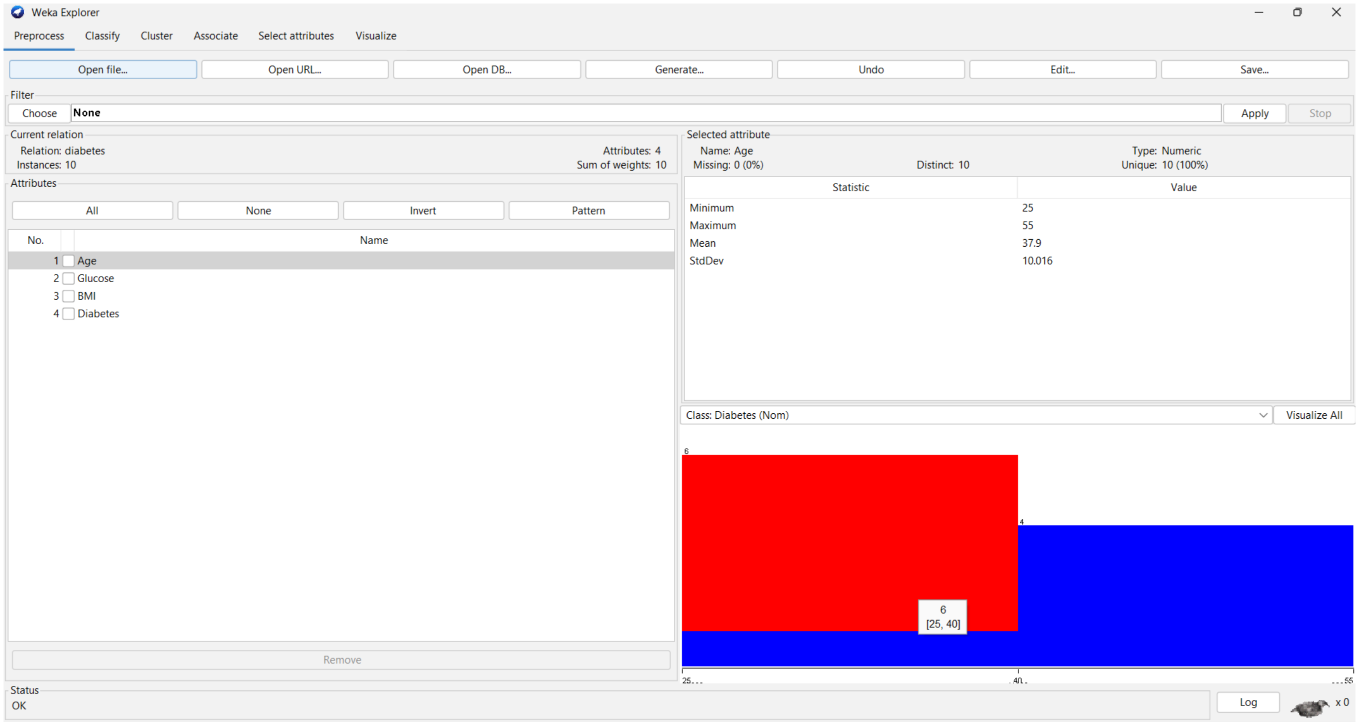


3. WEKA: Create patient Age, Glucose, BMI, Diabetes ARFF data; apply J48 and obtain confusion matrix.

Dataset: Diabetes

```
@relation diabetes
@attribute Age numeric
@attribute Glucose numeric
@attribute BMI numeric
@attribute Diabetes {Yes,No}
@data
25,95,22.5,No
32,130,28.1,Yes
41,145,31.2,Yes
29,105,24.8,No
50,160,34.5,Yes
36,120,27.0,No
45,150,32.1,Yes
27,100,23.5,No
55,170,36.0,Yes
39,115,26.5,No
```

Output:

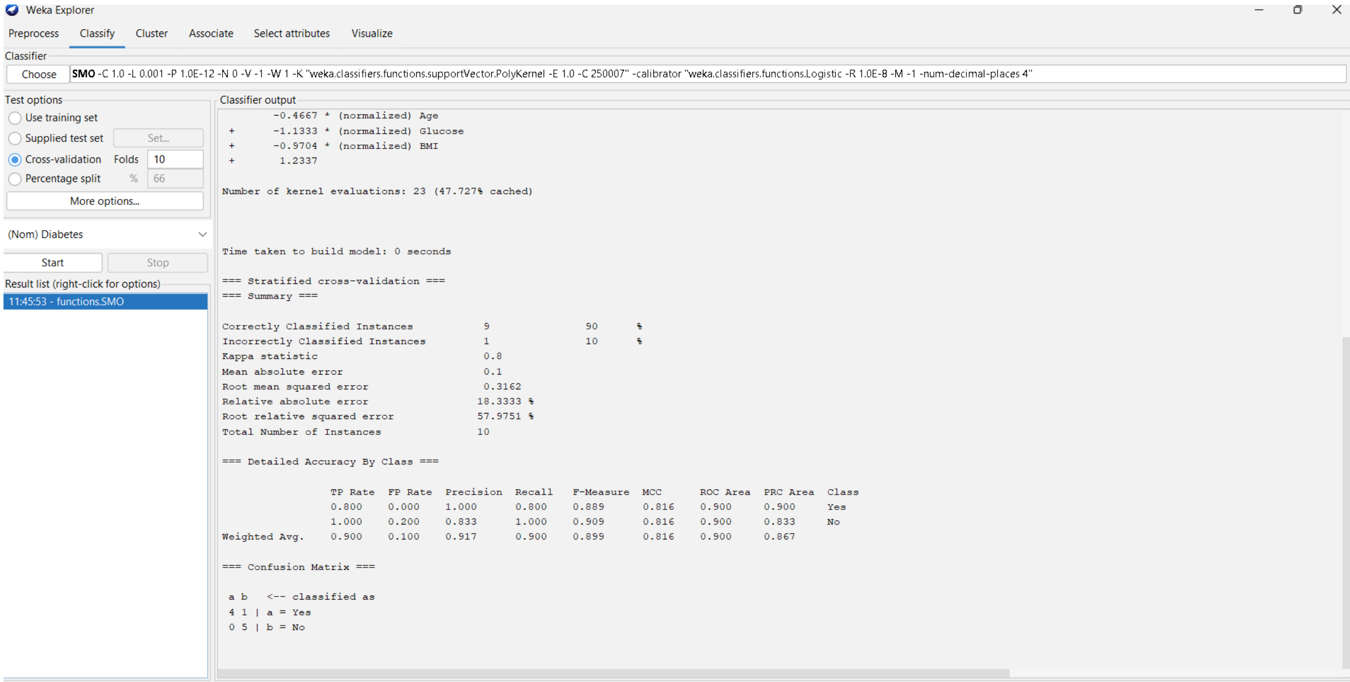


4. WEKA: Apply SMO to the diabetes data and calculate accuracy, precision, recall and F1; compare with J48.

Dataset: Diabetes – SMO

Use the same Diabetes ARFF dataset from Set 6 Q3.

Output:




```
       -0.4667 * (normalized) Age
 +     -1.1333 * (normalized) Glucose
 +     -0.9704 * (normalized) BMI
 +      1.2337

Number of kernel evaluations: 23 (47.727% cached)



Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances           9               90      %
Incorrectly Classified Instances         1               10      %
Kappa statistic                          0.8
Mean absolute error                      0.1
Root mean squared error                  0.3162
Relative absolute error                 18.3333 %
Root relative squared error             57.9751 %
Total Number of Instances               10

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.800    0.000    1.000      0.800    0.889      0.816    0.900     0.900     Yes
                 1.000    0.200    0.833      1.000    0.909      0.816    0.900     0.833     No
Weighted Avg.    0.900    0.100    0.917      0.900    0.899      0.816    0.900     0.867

=== Confusion Matrix ===

 a b   <-- classified as
 4 1 | a = Yes
 0 5 | b = No
```